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Solving initial value problems by differential quadrature method—Part 1: first-order equations

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SUMMARY

In this paper, the differential quadrature method is used to solve first-order initial value problems. The initial condition is given at the beginning of a time interval. The time derivative at a sampling grid point within the time interval can be expressed as a weighted linear sum of the given initial condition and the function values at the sampling grid points within the time interval. The order of accuracy and the stability property of the quadrature solutions depend on the locations of the sampling grid points. It is shown that the order of accuracy of the quadrature solutions at the end of a time interval can be improved to $2n-1$ or $2n$ if the n sampling grid points are chosen carefully. In fact, the approximate solutions are equivalent to the generalized Padé approximations. The resultant algorithms are therefore unconditionally stable with controllable numerical dissipation. The corresponding sampling grid points are found to be given by the roots of the modified shifted Legendre polynomials. From the numerical examples, the accuracy of the quadrature solutions obtained by using the proposed sampling grid points is found to be better than those obtained by the commonly used uniformly spaced or Chebyshev–Gauss–Lobatto sampling grid points. Copyright © 2001 John Wiley & Sons, Ltd.

KEY WORDS: single-step time-marching schemes; higher-order accurate algorithms; controllable numerical dissipation; non-linear transient analysis; ODE solver

1. INTRODUCTION

The differential quadrature method (DQM) is becoming a distinct numerical solution technique for the initial and/or boundary value problems of physical and engineering sciences [1]. The method is conceptually simple and the implementation is straightforward. It has been recognized that the differential quadrature method has the capability of producing highly accurate solutions with minimal computational effort [1, 2] when the method is applied to problems with globally smooth solutions. It has also been shown that the differential quadrature method is essentially equivalent to the general collocation method [3, 4] and the highest-order finite difference method [5].

The idea of quadrature rule for derivatives in the differential quadrature method was proposed by Bellman and Casti [6] in the early 1970s as an analogous extension of the quadrature for

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integrals. Later, Bellman *et al.* [7] applied the differential quadrature method to provide rapid and accurate solutions for initial-value non-linear partial differential equations. Since then, the differential quadrature method has been extended to tackle various engineering boundary value problems [e.g. 8–12]. A comprehensive review of the chronological development of the differential quadrature method can be found in Reference [1].

The differential quadrature method can be used for both spatial and temporal discretization [1, 13–15]. However, the method is usually used for spatial discretization only. The resultant ordinary differential equations in time can be solved by employing established time step integration schemes (for example, Newmark method or Runge–Kutta method). According to Bellman and Casti [6], the differential quadrature method was initially proposed 'to offer a new technique for the numerical solution of initial value problems for ordinary and partial differential equations with particular relevance to certain difficult questions arising from long-term integration'. For long-term integration, one of the main problems is numerical stability. A step-by-step time integration approach may have accumulation of evaluation. The round-off errors may seriously contaminate the solution, and may even obscure the actual value. However, not much work has been done in this area for the differential quadrature method.

In this paper, unconditionally stable higher-order accurate time step integration algorithms for first-order initial value problems based on the differential quadrature method are presented. Peters and Izadpanah [16] had pointed out that higher-order algorithms in time can be made competitive with conventional time-marching algorithms, particularly if high accuracy is needed (see also Reference [2]).

The manuscript is arranged as follows. In Section 2, the differential quadrature method and the methods to determine the weighting coefficients are briefly discussed. In Section 3, the differential quadrature rules for the initial value problems are presented. In Section 4, the accuracy of the quadrature solutions for linear problems is studied. The sampling grid points are chosen to optimize the order of accuracy of the quadrature solutions at the end of the time interval. The computation of the weighting coefficients is then discussed in Section 5. In Section 6, some lower-order algorithms for the first-order equations are presented. Numerical examples are given in Section 7 to verify the computational efficiency of the proposed sampling grid points. Conclusions are then given in Section 8.

2. DIFFERENTIAL QUADRATURE METHOD

In the differential quadrature method, the values of the derivatives at each sampling grid point are expressed as weighted linear sums of the function values at all sampling grid points within the domain under consideration, i.e.

$$\left.\frac{\mathrm{d}^r}{\mathrm{d}x^r}\Psi(x)\right|_{x=x_i} = \sum_{k=0}^{n} A_{ik}^{(r)}\Psi(x_k) \quad \text{for } i=0,1,2,\ldots,n \tag{1}$$

Equation (1) relates the rth derivative of the function $\Psi(x)$ at a sampling grid point $x=x_i$ to the function value $\Psi_k=\Psi(x_k)$ at $x=x_k$, with $A_{ik}^{(r)}$ denoting the corresponding weighting coefficient. The weighting coefficients can be determined such that Equation (1) is satisfied exactly for $n+1$ linearly independent test functions. These test functions (also called trial functions) can be polynomials (such as Lagrange, Jacobi, Legendre, Hermite or Chebyshev polynomials), trigonometric functions, sinc functions or spline functions.

The weighting coefficients can be written collectively in matrix form as

$$[\mathbf{A}^{(r)}]=\begin{bmatrix} A_{00}^{(r)} & A_{01}^{(r)} & \cdots & A_{0n}^{(r)} \\ A_{10}^{(r)} & A_{11}^{(r)} & \cdots & A_{1n}^{(r)} \\ \vdots & \vdots & & \vdots \\ A_{n0}^{(r)} & A_{n1}^{(r)} & \cdots & A_{nn}^{(r)} \end{bmatrix} \tag{2}$$

If the polynomials of degree up to n are used as the test functions (i.e. $\Psi(x)$ can be $1, x, x^2, \ldots,$ or x^n), it can be shown that $[\mathbf{A}^{(1)}]$ can be expressed as

$$[\mathbf{A}^{(1)}]=\begin{bmatrix} 1 & x_0 & \cdots & x_0^n \\ 1 & x_1 & \cdots & x_1^n \\ \vdots & \vdots & & \vdots \\ 1 & x_n & \cdots & x_n^n \end{bmatrix}\begin{bmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & n \\ & & & 0 \end{bmatrix}\begin{bmatrix} 1 & x_0 & \cdots & x_0^n \\ 1 & x_1 & \cdots & x_1^n \\ \vdots & \vdots & & \vdots \\ 1 & x_n & \cdots & x_n^n \end{bmatrix}^{-1}=[\mathbf{V}][\mathbf{\Gamma}][\mathbf{V}]^{-1} \tag{3}$$

where $x_0, x_1, \ldots, x_n$ are the co-ordinates of the $n+1$ sampling grid points. The $n+1$ sampling grid points are arbitrary but must all be distinct. Furthermore, $[\mathbf{A}^{(r)}]$ can be computed from $[\mathbf{A}^{(1)}]$ recursively as

$$[\mathbf{A}^{(r)}]=[\mathbf{A}^{(1)}]\ [\mathbf{A}^{(r-1)}]=[\mathbf{A}^{(r-1)}]\ [\mathbf{A}^{(1)}]=[\mathbf{V}][\mathbf{\Gamma}]^r[\mathbf{V}]^{-1} \quad \text{for } r>1 \tag{4}$$

Obviously, $[\mathbf{A}^{(r)}]=[\mathbf{0}]$ if $r \geqslant n+1$ as $[\mathbf{\Gamma}]^r=[\mathbf{0}]$ when $r \geqslant n+1$.

It can be seen that Equation (3) is inherently ill conditioned when n is large. The weighting coefficients calculated directly from Equation (3) may not be accurate if $n>20$ [1]. Alternatively, the weighting coefficients in $[\mathbf{A}^{(1)}]$, $[\mathbf{A}^{(2)}], \ldots, [\mathbf{A}^{(n)}]$ can also be evaluated explicitly as [3, 12, 17]

$$A_{ik}^{(1)}=\frac{\Pi(x_i)}{(x_i-x_k)\Pi(x_k)} \quad \text{for } i\neq k,\ 0\leqslant i,\ k\leqslant n \quad \text{where } \Pi(x_i)=\prod_{\substack{j=0\\ j\neq i}}^{n}(x_i-x_j) \tag{5a}$$

$$A_{ik}^{(r)}=r\left[A_{ii}^{(r-1)}A_{ik}^{(1)}-\frac{A_{ik}^{(r-1)}}{x_i-x_k}\right] \quad \text{for } i\neq k,\ 0\leqslant i,\ k\leqslant n,\ 2\leqslant r\leqslant n \tag{5b}$$

and

$$A_{ii}^{(r)}=-\sum_{\substack{k=0\\ k\neq i}}^{n}A_{ik}^{(r)} \quad \text{for } i=0,1,2,\ldots,n \quad \text{and} \quad 1\leqslant r\leqslant n \tag{5c}$$

2.1. Choices of sampling grid points

The accuracy of the quadrature solutions is dictated by the choice of the locations of the sampling grid points. However, the issue of the proper choice of the sampling grid points remains largely an unclear matter [1]. It is natural and convenient to choose sampling grid points with equal spacing, i.e.

$$x_i=\frac{i}{n}\Delta x \quad \text{for } i=0,1,2,\ldots,n \tag{6}$$

where Δx is the length of the domain interval under consideration. However, non-uniform grids (for example, zeros of the shifted Legendre [7] and Chebyshev [3, 18, 19] polynomials) have been demonstrated to enhance the accuracy of the quadrature solutions and can tackle the problem of deterioration of the quadrature solutions. It was also reported [1] that the Chebyshev–Gauss–Lobatto sampling grid points

$$x_i = \frac{1}{2}\left(1 - \cos\left(\frac{i\pi}{n}\right)\right)\Delta x \quad \text{for } i = 0, 1, 2, \ldots, n \tag{7}$$

performed consistently better than the equally spaced, Legendre and Chebyshev sampling grid points in a variety of problems.

In this paper, the sampling grid points are determined by optimizing the order of accuracy of the approximate solutions at the end of the time interval. As shown in the numerical examples, the proposed sampling grid points are found to be better in giving highly accurate quadrature solutions.

3. QUADRATURE RULES FOR INITIAL VALUE PROBLEMS

The main difference between initial value problems and boundary value problems is that the known boundary conditions are all given at the beginning of the domain (i.e. at the initial time of a time interval). The results at the other end of the time interval are in general unknown and are to be determined. To be consistent with the usual notation, the independent variable is denoted by t, instead of x.

A system of first-order equations can be written in the matrix form as

$$[\mathbf{C}]\frac{\mathrm{d}}{\mathrm{d}t}\{\mathbf{u}\} + [\mathbf{K}]\{\mathbf{u}\} = \{\mathbf{F}(t)\} \tag{8}$$

with initial condition $\{\mathbf{u}(t=0)\} = \{\mathbf{u}_0\}$. Equation (8) is parabolic. It can be used to describe the transient problems (such as transient heat conduction, soil consolidation, seepage through porous media, fluid and gas dynamics, mass transport, dynamics of stiffly plastic structures and many others) after employing some spatial discretization techniques. Besides, higher-order equations can be transformed into an equivalent system of first-order equations by introducing more independent variables.

The resultant system of equations from Equation (8) is coupled in general. For linear problems, Equation (8) can be uncoupled by using the modal decomposition method. It would be cumbersome and difficult to study the characteristics of a proposed algorithm by applying it to the coupled equations directly. It has been rigorously established that for linear problems, the integration of the uncoupled equations is equivalent to the integration of the original system [20–22]. It is therefore more convenient and sufficient for the purpose of investigating the characteristics of a proposed algorithm by considering the equation of motion of a single-degree-of-freedom system in the standard normalized form

$$\dot{y} = \lambda\Delta t y + f(\tau\Delta t)\Delta t \quad \text{with } y(\tau = 0) = y_0 \tag{9}$$

where $\tau = t/\Delta t$ is the non-dimensional time, Δt is the length of the time interval under consideration, λ may be complex and an over-dot is used to denote the differentiation with respect to τ.

The conclusions in stability and accuracy are valid for the original multi-degree-of-freedom system given in Equation (8) as well. Furthermore, the exact solution of Equation (9) can be written as

$$\bar{y}(\tau) = y_0 \exp(\lambda \Delta t \tau) + \int_0^\tau \exp(\lambda \Delta t(\tau - \bar{\tau})) f(\bar{\tau}\Delta t)\Delta t \, \mathrm{d}\bar{\tau} \tag{10}$$

In the following, the differential quadrature method is used to find an approximate solution over a time interval $0 \leqslant t \leqslant \Delta t$ (or $0 \leqslant \tau \leqslant 1$). In the spirit of differential quadrature, the derivatives and the function values are related to Equation (1). Since $\dot{y}(0)$ can be determined from Equation (9) as $\lambda \Delta t y_0 + f(0)\Delta t$, the first sampling grid point τ_0 can be assumed to be 0. In this case, the derivatives at the other n locations $\dot{y}_1 = \dot{y}(\tau_1), \ldots, \dot{y}_n = \dot{y}(\tau_n)$ are related to $y_0, y_1 = y(\tau_1), \ldots, y_n = y(\tau_n)$ by

$$\begin{Bmatrix} \dot{y}_1 \\ \vdots \\ \dot{y}_n \end{Bmatrix} = \begin{Bmatrix} A_{10}^{(1)} \\ \vdots \\ A_{n0}^{(1)} \end{Bmatrix} y_0 + \begin{bmatrix} A_{11}^{(1)} & \cdots & A_{1n}^{(1)} \\ \vdots & & \vdots \\ A_{n1}^{(1)} & \cdots & A_{nn}^{(1)} \end{bmatrix} \begin{Bmatrix} y_1 \\ \vdots \\ y_n \end{Bmatrix} \quad \text{or} \quad \{\dot{\mathbf{y}}\} = \{\mathbf{G}_0\} y_0 + [\mathbf{G}]\{\mathbf{y}\} \tag{11}$$

where

$$[\mathbf{G}] = \begin{bmatrix} 1 & \tau_1 & \cdots & \tau_1^{n-1} \\ 1 & \tau_2 & \cdots & \tau_2^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & \tau_n & \cdots & \tau_n^{n-1} \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 2 & & \\ & & \ddots & \\ & & & n \end{bmatrix} \begin{bmatrix} \tau_1 & \tau_1^2 & \cdots & \tau_1^n \\ \tau_2 & \tau_2^2 & \cdots & \tau_2^n \\ \vdots & \vdots & & \vdots \\ \tau_n & \tau_n^2 & \cdots & \tau_n^n \end{bmatrix}^{-1} \tag{12a}$$

$$\{\mathbf{G}_0\} = -[\mathbf{G}] \begin{Bmatrix} 1 \\ \vdots \\ 1 \end{Bmatrix} = -[\mathbf{G}]\{\mathbf{1}\}, \quad \{\dot{\mathbf{y}}\} = \begin{Bmatrix} \dot{y}_1 \\ \vdots \\ \dot{y}_n \end{Bmatrix}, \quad \{\mathbf{y}\} = \begin{Bmatrix} y_1 \\ \vdots \\ y_n \end{Bmatrix} \tag{12b}$$

and $0 < \tau_1, \tau_2, \ldots, \tau_n \leqslant 1$ are the co-ordinates of the sampling grid points. The sampling grid points should be all distinct.

In Equation (11), the initial condition is naturally incorporated into the differential quadrature rule. The n analog equations of the governing differential equation at the sampling grid points $\tau_1, \tau_2, \ldots, \tau_n$ are given by

$$\begin{Bmatrix} \dot{y}_1 \\ \vdots \\ \dot{y}_n \end{Bmatrix} - \lambda \Delta t \begin{Bmatrix} y_1 \\ \vdots \\ y_n \end{Bmatrix} = \begin{Bmatrix} f(\tau_1 \Delta t) \\ \vdots \\ f(\tau_n \Delta t) \end{Bmatrix} \Delta t \quad \text{or} \quad \{\dot{\mathbf{y}}\} - \lambda \Delta t \{y\} = \{\mathbf{f}\}\Delta t \tag{13}$$

Using Equation (11), Equation (13) can be written as

$$([\mathbf{G}] - \lambda \Delta t [\mathbf{I}])\{\mathbf{y}\} = \{\mathbf{f}\}\Delta t - y_0\{\mathbf{G}_0\} \tag{14}$$

Hence, the n unknowns $y_1, \ldots, y_n$ in $\{\mathbf{y}\}$ can be solved from Equation (14). The interpolated solution for $y(\tau)$ within the time interval is given by

$$y(\tau) = \sum_{k=0}^{n} L_k^n(\tau) y_k \tag{15}$$

where $L_k^n(\tau)$ is the nth order Lagrange polynomial and is given by

$$L_k^n(\tau) = \prod_{j=0, j\neq k}^{n} \frac{\tau - \tau_j}{\tau_k - \tau_j} \tag{16}$$

Note that $\tau_0 = 0$. The result at the end of the time interval can be evaluated as

$$y(\tau = 1) = \sum_{k=0}^{n} L_k^n(1) y_k \tag{17}$$

Using Equation (17), a time step integration algorithm is then constructed. To advance to the next time step, the results at the end of a time interval can be used as the initial conditions for the next time step. Hence, it is important to determine the results at the end of a time interval as accurate as possible. The solution characteristics, such as the order of accuracy and the numerical stability, of course depend on the sampling grid points. In general, uniformly spaced sampling grid points or the Chebyshev–Gauss–Lobatto points can be used in the calculations. In the next section, the sampling grid points are determined such that the accuracy of the numerical solutions at the end of the time interval is optimized.

4. ACCURACY OF THE QUADRATURE SOLUTIONS

To study the accuracy of the solution at the end of the time interval, $y(\tau)$ in Equation (15) is expanded into a polynomial of degree n as

$$y(\tau) = y_0 + Y_1\tau + Y_2\tau^2 + \cdots + Y_n\tau^n \tag{18}$$

where $Y_1, \ldots, Y_n$ are the polynomial coefficients and are related to $y_1, \ldots, y_n$ by

$$\begin{Bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{Bmatrix} = y_0 \begin{Bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{Bmatrix} + \begin{bmatrix} \tau_1 & \tau_1^2 & \cdots & \tau_1^n \\ \tau_2 & \tau_2^2 & \cdots & \tau_2^n \\ \vdots & \vdots & & \vdots \\ \tau_n & \tau_n^2 & \cdots & \tau_n^n \end{bmatrix} \begin{Bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{Bmatrix} \tag{19}$$

If $f(\tau\Delta t)\Delta t$ is similarly expressed as $F_0 + F_1\tau + F_2\tau^2 + \cdots + F_n\tau^n$, then

$$\{f\} = \begin{Bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{Bmatrix} = \begin{bmatrix} 1 & \tau_1 & \tau_1^2 & \cdots & \tau_1^n \\ 1 & \tau_2 & \tau_2^2 & \cdots & \tau_2^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \tau_n & \tau_n^2 & \cdots & \tau_n^n \end{bmatrix} \begin{Bmatrix} F_0 \\ F_1 \\ F_2 \\ \vdots \\ F_n \end{Bmatrix} \tag{20}$$

It can be shown that after some algebraic manipulations, Equation (14) can be written as

$$\left(\begin{bmatrix} 1 & & & \\ & 2 & & \\ & & \ddots & \\ & & & n \end{bmatrix} - \lambda\Delta t \begin{bmatrix} 0 & & & W_1 \\ 1 & \ddots & & W_2 \\ & \ddots & 0 & \vdots \\ & & 1 & W_n \end{bmatrix}\right) \begin{Bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{Bmatrix} = y_0 \lambda\Delta t \begin{Bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{Bmatrix}$$

$$+ \begin{bmatrix} 1 & 0 & & & W_1 \\ & 1 & \ddots & & W_2 \\ & & \ddots & 0 & \vdots \\ & & & 1 & W_n \end{bmatrix} \begin{Bmatrix} F_0 \\ F_1 \\ \vdots \\ F_n \end{Bmatrix} \tag{21}$$

where $W_1, W_2, \ldots, W_n$ are related to $\tau_1, \ldots, \tau_n$ by

$$\begin{Bmatrix} W_1 \\ W_2 \\ \vdots \\ W_n \end{Bmatrix} = \begin{bmatrix} 1 & \tau_1 & \cdots & \tau_1^{n-1} \\ 1 & \tau_2 & \cdots & \tau_2^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \tau_n & \cdots & \tau_n^{n-1} \end{bmatrix}^{-1} \begin{Bmatrix} \tau_1^n \\ \tau_2^n \\ \vdots \\ \tau_n^n \end{Bmatrix} \tag{22}$$

In other words, $W_1, W_2, \ldots, W_n$ are symmetric functions of $\tau_1, \ldots, \tau_n$. Alternatively, $\tau_1, \ldots, \tau_n$ can be regarded as the roots of the nth degree polynomial given by

$$\tau^n - W_1 - W_2\tau - W_3\tau^2 - \cdots - W_n\tau^{n-1} = 0 \tag{23}$$

It can be shown that if $f(\tau\Delta t) = 0$, the n unknowns $Y_1, \ldots, Y_n$ can be evaluated explicitly as

$$Y_j = \frac{(\lambda\Delta t)^j}{j!}\left(1 + \frac{1}{D}\sum_{k=n+1-j}^{n} (\lambda\Delta t)^k (n-k)!\, W_{n+1-k}\right) = \frac{(\lambda\Delta t)^j}{j!D}\left(n! - \sum_{k=1}^{n-j} (\lambda\Delta t)^k (n-k)!\, W_{n+1-k}\right) \tag{24}$$

where

$$D = n! - \sum_{k=1}^{n} (\lambda\Delta t)^k (n-k)!\, W_{n+1-k} \tag{25}$$

The solution characteristics of Equation (18) obtained by solving Equation (21) have been studied extensively by Fung [23–26]. In Reference [25], it is shown that if

$$W_k = \frac{(-1)^{n-k} n! n! (n+k-2)!}{(k-1)!(k-1)!(n+1-k)!(2n)!} \frac{2(n+\mu(k-1))}{1+\mu} \tag{26}$$

then the approximate solution given by Equation (18) (and hence Equation (15)) at the end of the time step ($t = \Delta t$ or $\tau = 1$) in the form

$$y(\tau = 1) = y_0 + Y_1 + \cdots + Y_n \tag{27}$$

is equivalent to the generalized Padé approximations. Furthermore, the algorithmic parameter μ in Equation (26) can be used to control the numerical dissipation of the algorithm. If $\mu = 1$, the algorithm is non-dissipative at the high-frequency regime. If $\mu = 0$, the algorithm is asymptotic annihilating, i.e. the very high-frequency responses are eliminated in one time step. Besides, the order of accuracy of the solution at the end of the time step is in general $2n-1$ if $0 \leqslant \mu < 1$ and $2n$ if $\mu = 1$ (see Reference [25] for a more detailed discussion).

It can be seen that in general $\tau = 1$ is not a root of Equation (23). As a result, the solutions at the end of the time interval have to be obtained by interpolation or extrapolation using Equation (15) or Equation (18). However, it can be shown that if $\mu = 0$, then

$$W_1 + W_2 + \cdots + W_n = 1 \tag{28}$$

and $\tau = 1$ is one of the roots of Equation (23). Hence, τ_n can be assigned as 1. As a result, the time step integration can be proceeded by using y_n as the initial condition for the next time step directly. Note that in this case, the algorithm would be asymptotic annihilating since $\mu = 0$.

5. COMPUTATION OF [G] AND $\{\mathbf{G}_0\}$

From Equation (11), it can be seen that

$$[\mathbf{G}] = \begin{bmatrix} A_{11}^{(1)} & A_{12}^{(1)} & \cdots & A_{1n}^{(1)} \\ A_{21}^{(1)} & A_{22}^{(1)} & \cdots & A_{2n}^{(1)} \\ \vdots & \vdots & & \vdots \\ A_{n1}^{(1)} & A_{n2}^{(1)} & \cdots & A_{nn}^{(1)} \end{bmatrix} \quad \text{and} \quad \{\mathbf{G}_0\} = \begin{Bmatrix} A_{10}^{(1)} \\ A_{20}^{(1)} \\ \vdots \\ A_{n0}^{(1)} \end{Bmatrix} \tag{29}$$

Since $\tau_0 = 0$ and $\tau_1, \ldots, \tau_n$ are given by the roots of the polynomial in Equation (23), instead of using Equation (5), $A_{ik}^{(1)}$ can be conveniently obtained from

$$A_{ik}^{(1)} = \frac{M'(x_i)}{(x_i - x_k)M'(x_k)} \quad \text{for } i \neq k, \ 0 \leqslant i, \ k \leqslant n \tag{30a}$$

and

$$A_{ii}^{(1)} = \frac{M''(x_i)}{2M'(x_i)} \quad \text{for } i = 0, 1, \ldots, n \tag{30b}$$

where M, M' and M'' are given by

$$M(\tau) = \tau^{n+1} - W_1\tau - W_2\tau^2 - W_3\tau^3 - \cdots - W_n\tau^n \tag{31a}$$

and

$$M'(\tau) = (n+1)\tau^n - W_1 - 2W_2\tau - 3W_3\tau^2 - \cdots - nW_n\tau^{n-1} \tag{31b}$$

$$M''(\tau) = n(n+1)\tau^{n-1} - 2W_2 - 6W_3\tau - \cdots - n(n-1)W_n\tau^{n-2} \tag{31c}$$

Note that the polynomial in Equation (23) with coefficients given by Equation (26) is different from the orthogonal polynomials commonly used for the determination of the sampling grid points. However, it can be shown that Equation (23) with coefficients given by Equation (26) is equivalent to

$$(1+\mu)P_n^*(\tau) - (1-\mu)P_{n-1}^*(\tau) = 0 \tag{32}$$

where $P_n^*(x)$ is the shifted Legendre polynomial of degree n and is defined as

$$P_n^*(x) = P_n(2x-1) \tag{33}$$

and $P_n(x)$ is the nth Legendre polynomial defined as

$$P_n(x) = \frac{1}{n!2^n}\frac{\mathrm{d}^n}{\mathrm{d}x^n}(x^2-1)^n \tag{34}$$

such that the Legendre polynomials are orthogonal over the interval $[-1, 1]$, i.e.

$$\int_{-1}^{1} P_m(x)P_n(x)\,\mathrm{d}x = 0 \quad \text{if } m \neq n \text{ and } 1 \text{ if } m = n \tag{35}$$

As a result, it can be seen that if $\mu = 1$, the sampling grid points $\tau_1, \ldots, \tau_n$ are the roots of $P_n^*(x)$ which are the same as the usual Gauss integration points used over the range $[0, 1]$.

Bellman *et al.* [7] employed the roots of the shifted Legendre polynomials as the sampling grid points and $A_{ii}^{(1)}$ in this case can be simplified to

$$A_{ii}^{(1)} = \frac{1-2x_i}{2x_i(x_i-1)} \quad \text{for } i = 1, \ldots, n. \tag{36}$$

Note that even though Bellman *et al.* [7] had proposed the use of the roots of the shifted Legendre polynomials as the sampling grid points, the present proposed sampling grid points are different from theirs as $\tau_0 = 0$ is included as one of the sampling grid points as well here.

6. LOWER-ORDER ALGORITHMS

In the following, some simple lower-order algorithms with $n = 1$ and 2 are considered. Without loss of generality, y_0 can be assumed to be 1 for linear first-order equations.

6.1. $n = 1$

When $n = 1$, Equation (23) reduces to

$$\tau - 1/(1+\mu) = 0 \quad \text{or } \tau_1 = 1/(1+\mu) \tag{37}$$

It can be shown that the quadrature solution given by Equation (15) or Equation (18) is

$$y(\tau) = 1 + \frac{(1+\mu)\lambda\Delta t}{1+\mu-\lambda\Delta t}\tau \tag{38}$$

and the truncation error for $y(\tau)$ at any time τ is given by

$$y(\tau) - \exp(\tau\lambda\Delta t) = \frac{1}{2}\frac{\tau(2-(1+\mu)\tau)}{1+\mu}\lambda^2\Delta t^2 + O(\Delta t^3) \tag{39}$$

Hence, it can be seen that the quadrature solution is first-order accurate only in general except at $\tau = 2/(1+\mu) = 2\tau_1$, which is second-order accurate. As a result, if the result at the end of a time interval is to be of higher-order accuracy, τ_1 should be set as $\frac{1}{2}$ (and hence $\mu = 1$). In this case, the truncation error for $y(\tau = 1)$ is

$$y(\tau = 1) - \exp(\lambda \Delta t) = \tfrac{1}{12}\lambda^3 \Delta t^3 + O(\Delta t^4) \tag{40}$$

The algorithm is in fact equivalent to the well-known mid-point rule.

The stability of the algorithm can be directly related to the co-ordinate of the sampling grid point τ_1 as well. From Equation (38), the amplification factor is given by $y(\tau = 1)$ as

$$y(\tau = 1) = 1 + \frac{(1+\mu)\lambda \Delta t}{1 + \mu - \lambda \Delta t} = 1 + \frac{\lambda \Delta t}{1 - \tau_1 \lambda \Delta t} \tag{41}$$

It can be shown that if $0 \leqslant \tau_1 < \frac{1}{2}$ (hence $\mu > 1$), the algorithm is only conditionally stable since $|y(\tau = 1)| > 1$ for some $\lambda \Delta t$ if the real part of λ is non-positive. If $1 \geqslant \tau_1 \geqslant \frac{1}{2}$ (hence $0 \leqslant \mu \leqslant 1$), the algorithm is unconditionally stable since $|y(\tau = 1)| \leqslant 1$ for all $\lambda \Delta t$ if the real part of λ is non-positive.

In the conventional approach, τ_1 is equal to 1. This is equivalent to $\mu = 0$. As a result, the algorithm is first-order accurate only at the end of the time interval besides being asymptotically annihilating. This algorithm is also equivalent to the backward Euler method. Other values of τ_1 would give first-order accurate algorithms with the same computational effort and are therefore not very interesting.

6.2. $n = 2$

When $n = 2$, Equation (23) becomes

$$3(1+\mu)\tau^2 - 2(2+\mu)\tau + 1 = 0 \tag{42}$$

and the two roots τ_1 and τ_2 can be evaluated explicitly as

$$\tau_1 = \frac{2 + \mu - \sqrt{1 + \mu + \mu^2}}{3(1+\mu)} \quad \text{and} \quad \tau_2 = \frac{2 + \mu + \sqrt{1 + \mu + \mu^2}}{3(1+\mu)} \tag{43}$$

It can be shown that the weighting coefficient matrix $[\mathbf{G}]$ and vector $\{\mathbf{G}_0\}$ in Equation (12) are given by

$$[\mathbf{G}] = \begin{bmatrix} \dfrac{1}{2}\left(\dfrac{2\mu^2 - \mu - 1}{\sqrt{1+\mu+\mu^2}} + 4 + 2\mu\right) & -\dfrac{1}{2}\left(\dfrac{-2\mu^2 - 5\mu - 5}{\sqrt{1+\mu+\mu^2}} + 4 + 2\mu\right) \\ -\dfrac{1}{2}\left(\dfrac{2\mu^2 + 5\mu + 5}{\sqrt{1+\mu+\mu^2}} + 4 + 2\mu\right) & \dfrac{1}{2}\left(\dfrac{-2\mu^2 + \mu + 1}{\sqrt{1+\mu+\mu^2}} + 4 + 2\mu\right) \end{bmatrix} \tag{44a}$$

and

$$\{\mathbf{G}_0\} = 2\sqrt{1 + \mu + \mu^2} \begin{Bmatrix} -1 \\ 1 \end{Bmatrix} \tag{44b}$$

It can be shown that the quadrature solution can be written as

$$y(\tau)=1+\frac{-2((2+\mu)\lambda\Delta t-3(1+\mu))\lambda\Delta t\tau+3(1+\mu)\lambda^2\Delta t^2\tau^2}{\lambda^2\Delta t^2-2(2+\mu)\lambda\Delta t+6(1+\mu)} \tag{45}$$

and the truncation error for $y(\tau)$ at any time τ is given by

$$y(\tau)-\exp(\tau\lambda\Delta t)=\frac{1}{6}\frac{\tau(1-\tau)((1+\mu)\tau-1)}{1+\mu}\lambda^3\Delta t^3+O(\Delta t^4) \tag{46}$$

Hence, it can be seen that the quadrature solution is second-order accurate only in general. The order of accuracy is improved at the end of the time interval with $\tau=1$. In this case, Equation (45) reduces to

$$y(\tau=1)=\frac{\mu\lambda^2\Delta t^2+2(1+2\mu)\lambda\Delta t+6(1+\mu)}{\lambda^2\Delta t^2-2(2+\mu)\lambda\Delta t+6(1+\mu)} \tag{47}$$

and the truncation error in Equation (46) for $y(\tau=1)$ is given by

$$y(\tau=1)-\exp(\lambda\Delta t)=-\frac{1-\mu}{72(1+\mu)}\lambda^4\Delta t^4+O(\Delta t^5) \tag{48}$$

If μ is further chosen as 1, the accuracy of the quadrature solution at the end of the time interval can be further improved to

$$y(\tau=1)-\exp(\lambda\Delta t)=-\frac{1}{720}\lambda^5\Delta t^5+O(\Delta t^6) \tag{49}$$

and the quadrature solution is fourth-order accurate. Hence, it can be seen that if $\mu=0$,

$$\tau_1=\tfrac{1}{3} \quad \text{and} \quad \tau_2=1 \tag{50}$$

the algorithm is third-order accurate and asymptotic annihilating. If $\mu=1$,

$$\tau_1=\frac{1}{2}-\frac{\sqrt{3}}{6} \quad \text{and} \quad \tau_2=\frac{1}{2}+\frac{\sqrt{3}}{6} \tag{51}$$

and the algorithm is fourth-order accurate and non-dissipative. For any value of μ between -1 and 1, the resultant algorithm is third-order accurate and unconditionally stable.

In the conventional approach, the ends are usually included. Assume $\tau_0=0$, $\tau_1=\beta$ and $\tau_2=1$. It can be shown that the quadrature solution is given by

$$y(\tau)=1-\frac{((1+\beta)\lambda\Delta t-2)\lambda\Delta t\tau-\lambda^2\Delta t^2\tau^2}{\beta\lambda^2\Delta t^2-(1+\beta)\lambda\Delta t+2} \tag{52}$$

and the truncation error for $y(\tau)$ in Equation (52) is given by

$$y(\tau)-\exp(\tau\lambda\Delta t)=-\tfrac{1}{12}\tau(2\tau^2-3(\beta+1)\tau+6\beta)\lambda^3\Delta t^3+O(\Delta t^4) \tag{53}$$

From Equation (53), it can be seen that the quadrature solution is second-order accurate only in general. The order of accuracy at the end of the time interval with $\tau=1$ cannot be improved unless $\beta=\frac{1}{3}$. In this case, the algorithm is equivalent to the present algorithm given by Equation (50). In the conventional approach, $\beta=\frac{1}{2}$. The resultant algorithm is therefore only second-order accurate. Furthermore, if $\lambda<0$, it can be seen that $|y(\tau=1)|\leqslant 1$ for all $\lambda\Delta t$. The algorithms are therefore at least A_0 stable.

7. NUMERICAL EXAMPLES

In this section, three numerical examples are included to demonstrate the accuracy of the proposed sampling grid points in solving linear and non-linear first-order equations.

7.1. Example 1: algorithms with $n=3$ and $\mu=0$

It is difficult to analyse the algorithm characteristics with $n=3$ and general μ. Consider $\mu=0$. The polynomial given by Equation (23) to generate τ_1, τ_2 and τ_3 is

$$\tau^3 - \frac{9}{5}\tau^2 + \frac{9}{10}\tau - \frac{1}{10} = 0 \tag{54}$$

The three roots τ_1, τ_2 and τ_3 can be evaluated explicitly as

$$\tau_1 = \frac{2}{5} - \frac{\sqrt{6}}{10}, \quad \tau_2 = \frac{2}{5} + \frac{\sqrt{6}}{10}, \quad \tau_3 = 1 \tag{55}$$

The weighting coefficient matrix $[\mathbf{G}]$ and vector $\{\mathbf{G}_0\}$ in Equation (29) are given by

$$[\mathbf{G}] = \begin{bmatrix} 2+\dfrac{\sqrt{6}}{2} & \dfrac{29}{30}\sqrt{6}-\dfrac{6}{5} & -\dfrac{4}{15}\sqrt{6}+\dfrac{2}{5} \\ -\dfrac{29}{30}\sqrt{6}-\dfrac{6}{5} & 2-\dfrac{\sqrt{6}}{2} & \dfrac{4}{15}\sqrt{6}+\dfrac{2}{5} \\ \dfrac{8}{3}\sqrt{6}-1 & -\dfrac{8}{3}\sqrt{6}-1 & 5 \end{bmatrix} \quad \text{and} \quad \{\mathbf{G}_0\} = \begin{Bmatrix} -\dfrac{6}{5}(1+\sqrt{6}) \\ -\dfrac{6}{5}(1-\sqrt{6}) \\ -3 \end{Bmatrix} \tag{56}$$

It can be shown that the truncation errors for the quadrature solutions at the sampling grid points τ_1, τ_2 and τ_3 are given by

$$y_1 - \exp(\tau_1\lambda\Delta t) = \left(\frac{3}{20000} + \frac{\sqrt{6}}{2500}\right)\lambda^4\Delta t^4 + O(\Delta t^5) \tag{57a}$$

$$y_2 - \exp(\tau_2\lambda\Delta t) = \left(\frac{3}{20000} - \frac{\sqrt{6}}{2500}\right)\lambda^4\Delta t^4 + O(\Delta t^5) \tag{57b}$$

$$y_3 - \exp(\tau_3\lambda\Delta t) = \frac{1}{7200}\lambda^6\Delta t^6 + O(\Delta t^7) \tag{57c}$$

It can be seen from Equation (57c) that the solution at the end of the time interval (i.e. y_3) is fifth-order accurate. In general, the truncation error for the interpolated solution is given by

$$y(\tau) - \exp(\tau\lambda\Delta t) = -\frac{1}{120}\tau(-2+5\tau)(\tau-1)^2\lambda^4\Delta t^4 + O(\Delta t^5) \tag{58}$$

It can be seen that the algorithm is in general third-order accurate.

If uniformly spaced sampling grid points are used (i.e. $\tau_0=0, \tau_1=\frac{1}{3}, \tau_2=\frac{2}{3}, \tau_3=1$), it can be shown that the truncation error for the interpolated solution is given by

$$y(\tau) - \exp(\tau\lambda\Delta t) = -\frac{1}{216}\tau(3\tau-4)(3\tau^2-4\tau+2)\lambda^4\Delta t^4 + O(\Delta t^5) \tag{59}$$

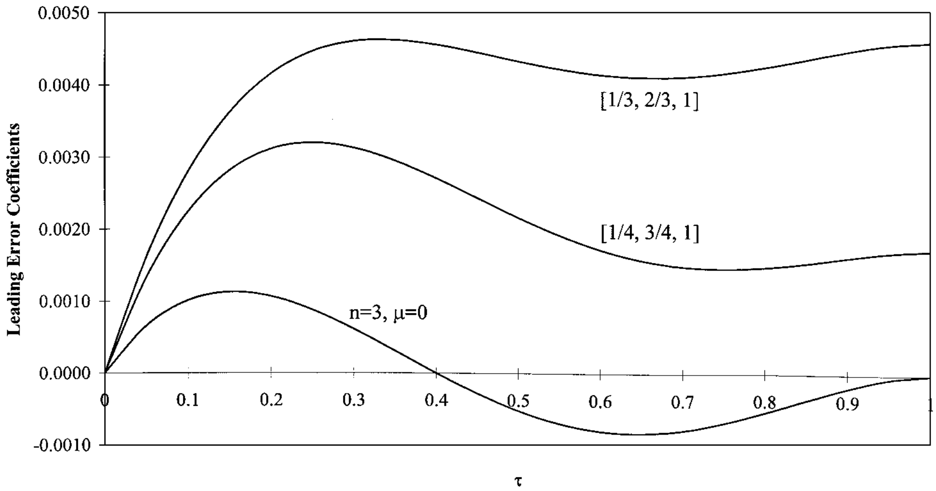

Figure 1. Magnitude of the leading error coefficients.

Similarly, if the Chebyshev–Gauss–Lobatto points are used (i.e. $\tau_0 = 0, \tau_1 = \frac{1}{4}, \tau_2 = \frac{3}{4}, \tau_3 = 1$), it can be shown that the truncation error for the interpolated solution is given by

$$y(\tau) - \exp(\tau\lambda\Delta t) = -\frac{1}{576}\tau(24\tau^3 - 64\tau^2 + 57\tau - 18)\lambda^4\Delta t^4 + O(\Delta t^5) \tag{60}$$

It can be seen that the three quadrature solutions in Equations (58)–(60) are all third-order accurate. The coefficients of $\lambda^4\Delta t^4$ in Equations (58)–(60) are however quite different and are plotted in Figure 1. It can be seen that the leading error term in Equation (58) is the smallest when $0 \leqslant \tau \leqslant 1$. Hence, it is reasonable to expect the present proposed sampling grid points to give better results than the uniformly spaced or Chebyshev–Gauss–Lobatto sampling grid points. The proposed sampling grid points have a further advantage that the quadrature solution at the end of the time interval is fifth-order accurate.

7.2. Example 2: cooling/heating by combined convection and radiation

Consider a lumped capacity model of a body subjected to heat transfer by combined convection and radiation. The time rate of change in temperature of the body is given by

$$\frac{d\Theta}{dt} = -(\Theta - \Theta_a) - \frac{F\sigma}{h}(\Theta^4 - \Theta_s^4) \tag{61}$$

where $\Theta = \Theta(t)$ is the absolute temperature of the body and t is a non-dimensional time. Besides, the numerical data provided by Lienhard [27] for a black sphere in an enclosure (i.e. $\Theta_a = 293$ K, $\Theta_s = 1000$ K, $h = 300$ W/m^2 K, $F = 1$, $\sigma = 5.6697 \times 10^{-8}$ W/m^2 K^4) are used. It can be calculated that the steady state or final temperature Θ_f is 472.565 K, which is independent of the initial temperature.

It is of interest to find the temperature versus time history of the body for a given initial temperature. In Reference [1], the problem domain is chosen to be $0 \leqslant t \leqslant t_f$, where t_f is a sufficiently

Table I. Numerical solutions for Example 2 by various methods.

Method/grid points	Δt	Dimensionless time, t: 1	2	3	4	15
		$\Theta_0 = 1000$ K, $\Theta(\tau)$ by various methods in K				
$[0, 1/2, 1]$	1	638.15	528.63	491.89	479.26	472.56
$n = 2,\ \mu = 0$	1	626.80	522.63	489.16	478.10	472.56
$n = 2,\ \mu = 1$	1	631.98	525.34	490.38	478.61	472.56
$[0, 1/3, 2/3, 1]$	1	633.71	526.05	490.65	478.71	472.56
$[0, 1/4, 3/4, 1]$	1	632.03	525.29	490.34	478.59	472.56
$n = 3,\ \mu = 0$	1	631.51	525.07	490.25	478.56	472.56
$n = 3,\ \mu = 1$	1	631.42	525.03	490.23	478.55	472.56
$n = 3,\ \mu = 1$	2	626.41	524.73	489.92	478.50	472.56
$[0, 1/4, 1/2, 3/4, 1]$	2	634.80	525.86	490.60	478.66	472.56
$[0, 0.146, 0.5, 0.854, 1]$	2	631.21	525.41	490.35	478.60	472.56
$n = 4,\ \mu = 1$	2	631.66	525.04	490.24	478.55	472.56
DQM, $N = 26$ [1]	15	631.43	525.03	490.24	478.55	472.57

large time so that the final temperature Θ_f is reached. The differential quadrature rules were then applied to the sample grid points within the whole time domain $0 \leqslant t \leqslant t_f$.

In the following, the time step integration approach is adopted. The time domain is studied using a time step size of Δt. The differential quadrature method is applied over the time interval Δt only. The results at the end of a time interval will be used as the initial conditions for the next time interval. The iterative procedure presented in Reference [1] is adopted here to solve the non-linear problem.

Consider an initial temperature $\Theta_0 = 1000$ K. The results are given in Table I. Intermediate numerical results (for example, $t = 1$ when $\Delta t = 2$) are obtained by Lagrange interpolation. The tabulated results demonstrate that for the same computational effort, the present algorithms give better numerical solutions than the conventional algorithms using the uniformly spaced sampling grid points or the Chebyshev–Gauss–Lobatto points.

7.3. Example 3: differential equations with variable coefficients

The differential quadrature method can be used to tackle differential equations with variable coefficients easily. Consider the following first-order differential equation:

$$\frac{dy(t)}{dt} = \cos(t)y(t) \tag{62}$$

with initial condition $y(0) = 1$. The exact solution is obviously given by

$$y_e(t) = \exp(\sin(t)) \tag{63}$$

It can be seen that the exact solution is periodic with a period $T = 2\pi$. The exact solution also varies only from e^{-1} to e. If the differential quadrature method is used to solve the problem, y_i at $t = \tau_i \Delta t$ can be obtained by solving

$$([\mathbf{G}] - [\Lambda]\Delta t)\{\mathbf{y}\} = -\{\mathbf{G}_0\}y_0 \tag{64}$$

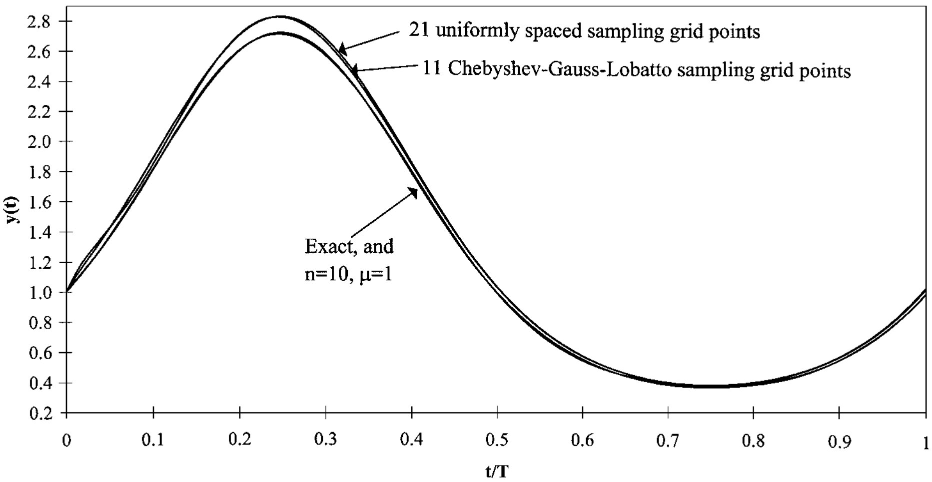

Figure 2. Numerical results for Example 3 by various methods.

where $[\Lambda]$ is a diagonal matrix

$$[\Lambda] = \begin{bmatrix} \cos\left(t_0 + \dfrac{\Delta t}{n}\right) & & & \\ & \ddots & & \\ & & \cos\left(t_0 + \dfrac{(n-1)\Delta t}{n}\right) & \\ & & & \cos(t_0 + \Delta t) \end{bmatrix} \tag{65}$$

Δt is the time step, t_0 and y_0 are the initial time and initial displacement, respectively, at the beginning of the time interval under consideration. The interpolated solution is then given by Equation (15). In the time step integration algorithms, the solution at the end of a time interval can be used as the initial condition for the next time interval and the solution over the entire time domain can then be studied.

Since the solution is periodic, consider the solution over one period with $0 \leqslant t \leqslant 2\pi$. If the whole range is studied by the differential quadrature method, there is only one time step with $\Delta t = 2\pi$, $t_0 = 0$ and $y_0 = 1$. The quadrature solutions given by various methods are shown in Figure 2. It can be seen that the present algorithm with $n = 10$ and $\mu = 1$ gives very accurate quadrature solutions. With the same computational effort, the differential quadrature method using 11 Chebyshev–Gauss–Lobatto sampling grid points does not give accurate results. More sampling grid points are required for uniformly spaced sampling grid points in order to generate comparable quadrature solutions. Hence, it is verified that the proposed sampling grid points can be used to generate highly accurate solutions.

If the time step integration strategy is adopted, lower-order differential quadrature rules can be used within a time interval. Consider various algorithms with $n = 2, 3$ and 4 (i.e. with 3, 4 and 5 sampling grid points after including the initial point $\tau_0 = 0$). Figure 3 shows the maximum relative errors of the quadrature solutions over one period against the number of time steps used over one

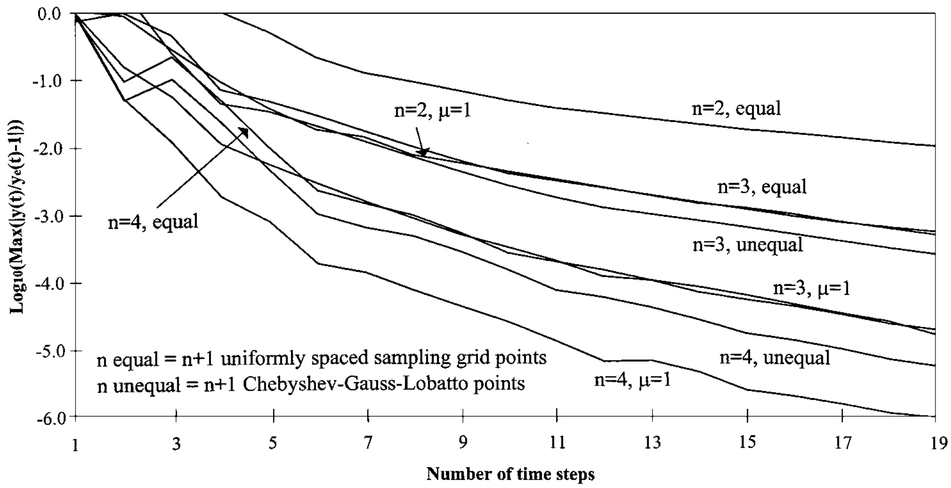

Figure 3. Errors in the numerical solutions given by various methods in Example 3.

period. Log_{10} of the relative errors would give an indication of the number of significant figures in the quadrature solutions. It can be seen that for the same computational effort (i.e. the same number of sampling grid points), the algorithms using the proposed sampling grid points give much better accuracy.

8. CONCLUSIONS

In this paper, unconditionally stable higher-order accurate time step integration algorithms suitable for first-order differential equations based on the differential quadrature method are presented. The numerical dissipation in the algorithm is a directly controllable algorithmic parameter. The required sampling grid points to generate unconditionally stable higher-order accurate time step integration algorithms are given by the roots of a polynomial in terms of numerical dissipation control parameter μ. If n sampling grid points are used, the interpolated solutions are in general at least nth-order accurate. By using the proposed sampling grid points, the order of accuracy can be improved to $2n-1$ or $2n$ for the quadrature solutions given at the end of the time interval. The approximate solutions are in fact equivalent to the generalized Padé approximations. Numerical examples are presented to verify the computational efficiency of the proposed sampling grid points.

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